MINISTRY OF EDUCATION
IMAM ABDULRAHMAN BIN FAISAL UNIVERSITY
COLLEGE OF COMPUTER SCIENCE & INFORMATION TECHNOLOGY

وزارة التعليم
جامعة الإمام
عبدالرحمن بن فيصل
كلية علوم الحاسب
وتقنية المعلومات

جامعة الإمام عبدالرحمن بن فيصل
IMAM ABDULRAHMAN BIN FAISAL UNIVERSITY

FUNDAMENTALS OF COMMUNICATIONS

PROJECT REPORT

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Introduction

In this report, we present the steps taken to design and implement our network topology using Cisco Packet Tracer. The objective was to create a fully functional enterprise network with proper IP addressing, routing, DHCP configuration, VLANs, and secure wireless connectivity. Our design is based on two main routers (Router 1 and Router 2) interconnected via serial links.

Topology

Our network topology consists of:

- Two Routers (Router 1 and Router 2) configured with static IP addresses and connected via serial links for WAN communication.
- Multiple Switches: Switch 1 connected to Router 1 and Switch 2 connected to Router 2.
- PCs and Printers: Connected to the respective switches for each router.
- DHCP Server: Configured to provide dynamic IP addressing for certain devices.
- Access Point: Providing secure wireless connectivity for mobile devices.
- IP Phones: Configured and connected via switches for VoIP functionality.

Video simulation
PacketTracer File

Table of addresses

Device	Interface	IP Address	Subnet Mask	Default Gateway
Router 1	Ge 0/0	10.10.10.30	255.255.255.252	N/A
	Ge 0/1	10.10.10.45	255.255.255.252	N/A
	Ge 0/2	10.10.10.53	255.255.255.252	N/A
	Se 0/2/0	10.10.10.26	255.255.255.252	N/A
	Se 0/2/1	10.10.10.38	255.255.255.252	N/A
Router 2	Ge 0/0	10.10.10.22	255.255.255.252	N/A
	Ge 0/1	10.10.10.41	255.255.255.252	N/A
	Ge 0/2	10.10.10.49	255.255.255.252	N/A
	Se 0/2/0	10.10.10.34	255.255.255.252	N/A

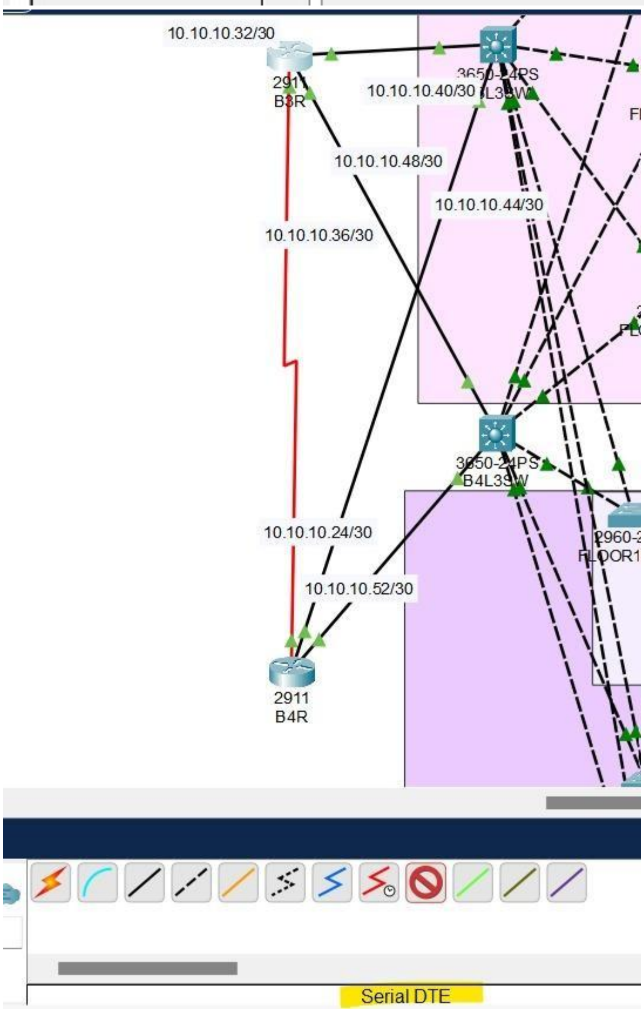
	Se 0/2/1	10.10.10.37	255.255.255.252	N/A
Switch 0	Ge 1/0/1 – Ge 1/0/24 Ge 1/1 – Ge 1/4	N/A	N/A	N/A
Switch 1	Ge 1/0/1 – Ge 1/0/24 Ge 1/1/1 – Ge 1/1/4	N/A	N/A	N/A
Switch 2	Ge 1/0/1 – Ge 1/0/24 Ge 1/1/1 – Ge 1/1/2	N/A	N/A	N/A
Switch 3	Fa 0/1 – Fa 0/24 Ge 0/1 – Ge 0/2	N/A	N/A	N/A
Switch 4	Fa 0/1 – Fa 0/24 Ge 0/1 – Ge 0/2	N/A	N/A	N/A
Switch 5	Fa0/1 - Fa0/24 Ge 0/1 – Ge 0/2	N/A	N/A	N/A
PC0	FastEthernet0	192.168.11.140	255.255.255.192	192.168.11.129
PC1	FastEthernet0	192.168.11.145	255.255.255.192	192.168.11.129
PC2	FastEthernet0	192.168.11.143	255.255.255.192	192.168.11.129
PC3	FastEthernet0	192.168.11.147	255.255.255.192	192.168.11.129
PC4	FastEthernet0	192.168.11.149	255.255.255.192	192.168.11.129
PC5	FastEthernet0	192.168.11.137	255.255.255.192	192.168.11.129
PC6	FastEthernet0	192.168.11.141	255.255.255.192	192.168.11.129
PC7	FastEthernet0	192.168.11.144	255.255.255.192	192.168.11.129
PC8	FastEthernet0	192.168.11.135	255.255.255.192	192.168.11.129
PC9	FastEthernet0	192.168.11.148	255.255.255.192	192.168.11.129
PC10	FastEthernet0	192.168.11.146	255.255.255.192	192.168.11.129
PC11	FastEthernet0	192.168.11.201	255.255.255.192	192.168.11.193
PC12	FastEthernet0	192.168.11.206	255.255.255.192	192.168.11.193
PC13	FastEthernet0	192.168.11.209	255.255.255.192	192.168.11.193
PC14	FastEthernet0	192.168.11.210	255.255.255.192	192.168.11.193
PC15	FastEthernet0	192.168.11.208	255.255.255.192	192.168.11.193
PC16	FastEthernet0	192.168.11.218	255.255.255.192	192.168.11.193
PC17	FastEthernet0	192.168.11.211	255.255.255.192	192.168.11.193
PC18	FastEthernet0	192.168.11.220	255.255.255.192	192.168.11.193
PC19	FastEthernet0	192.168.11.212	255.255.255.192	192.168.11.193
PC20	FastEthernet0	192.168.11.205	255.255.255.192	192.168.11.193
PC21	FastEthernet0	192.168.12.17	255.255.255.192	192.168.12.1
PC22	FastEthernet0	192.168.12.21	255.255.255.192	192.168.12.1
PC23	FastEthernet0	192.168.12.14	255.255.255.192	192.168.12.1
PC24	FastEthernet0	192.168.12.10	255.255.255.192	192.168.12.1
PC25	FastEthernet0	192.168.12.8	255.255.255.192	192.168.12.1
PC26	FastEthernet0	192.168.12.22	255.255.255.192	192.168.12.1
PC27	FastEthernet0	192.168.12.23	255.255.255.192	192.168.12.1
PC28	FastEthernet0	192.168.12.24	255.255.255.192	192.168.12.1
PC29	FastEthernet0	192.168.12.18	255.255.255.192	192.168.12.1
PC30	FastEthernet0	192.168.12.16	255.255.255.192	192.168.12.1
PC31	FastEthernet0	192.168.12.7	255.255.255.192	192.168.12.1

PC32	FastEthernet0	192.168.12.10	255.255.255.192	192.168.12.1
PC33	FastEthernet0	192.168.12.77	255.255.255.192	192.168.12.65
PC34	FastEthernet0	192.168.12.81	255.255.255.192	192.168.12.65
PC35	FastEthernet0	192.168.12.87	255.255.255.192	192.168.12.65
PC36	FastEthernet0	192.168.12.84	255.255.255.192	192.168.12.65
PC37	FastEthernet0	192.168.12.91	255.255.255.192	192.168.12.65
PC38	FastEthernet0	192.168.12.71	255.255.255.192	192.168.12.65
PC39	FastEthernet0	192.168.12.78	255.255.255.192	192.168.12.65
PC40	FastEthernet0	192.168.12.90	255.255.255.192	192.168.12.65
PC41	FastEthernet0	192.168.12.80	255.255.255.192	192.168.12.65
PC42	FastEthernet0	192.168.12.72	255.255.255.192	192.168.12.65
PC43	FastEthernet0	192.168.12.141	255.255.255.192	192.168.12.129
PC44	FastEthernet0	192.168.12.157	255.255.255.192	192.168.12.129
PC45	FastEthernet0	192.168.12.156	255.255.255.192	192.168.12.129
PC46	FastEthernet0	192.168.12.140	255.255.255.192	192.168.12.129
PC47	FastEthernet0	192.168.12.142	255.255.255.192	192.168.12.129
PC48	FastEthernet0	192.168.12.139	255.255.255.192	192.168.12.129
PC49	FastEthernet0	192.168.12.137	255.255.255.192	192.168.12.129
PC50	FastEthernet0	192.168.12.138	255.255.255.192	192.168.12.129
PC51	FastEthernet0	192.168.12.154	255.255.255.192	192.168.12.129
PC52	FastEthernet0	192.168.12.143	255.255.255.192	192.168.12.129
PC53	FastEthernet0	192.168.12.194	255.255.255.192	0.0.0.0
Laptop1	Wireless0	192.168.12.145	255.255.255.192	192.168.12.129
Laptop2	Wireless0	192.168.12.147	255.255.255.192	192.168.12.129
Laptop3	Wireless0	192.168.12.144	255.255.255.192	192.168.12.129
Printer1	FastEthernet0	192.168.11.142	255.255.255.192	192.168.11.129
Printer2	FastEthernet0	192.168.11.202	255.255.255.192	192.168.11.193
Printer3	FastEthernet0	192.168.12.11	255.255.255.192	192.168.12.1
Printer4	FastEthernet0	192.168.12.88	255.255.255.192	192.168.12.65
Printer5	FastEthernet0	192.168.12.149	255.255.255.192	192.168.12.129
Server1	FastEthernet0	192.168.12.197	255.255.255.192	192.168.12.193
Server2	FastEthernet0	192.168.12.196	255.255.255.192	192.168.12.193
Server3	FastEthernet0	192.168.12.198	255.255.255.192	192.168.12.193
IP Phone 1				

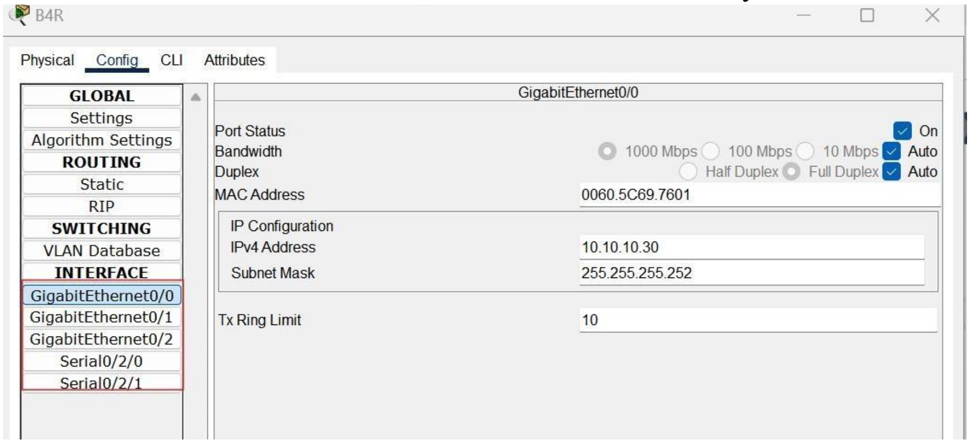
Step-by-Step Implementation

1. Step 1: Installed HWIC-2T modules on both routers for serial connectivity. Used serial DCE cables between Router 1 and Router 2.


B3R
Physical
Config
CLI
Attributes
MODULES
HWIC-1GE-SFP
HWIC-2T
HWIC-4ESW
HWIC-8A
WIC-Cover
GLC-LH-SMD
Physical Device View
Zoom In
Original Size


10.10.10.32/30
2911
B3R
3650-24PS
10.10.10.40/30
10.10.10.48/30
10.10.10.44/30
10.10.10.36/30
3650-24PS
B4L3SW
10.10.10.24/30
10.10.10.52/30
2911
B4R
Serial DTE

2. Step 2: Configured static IP addresses on Router 1 and Router 2. Assigned IP addresses to the router interfaces, and verified connectivity.



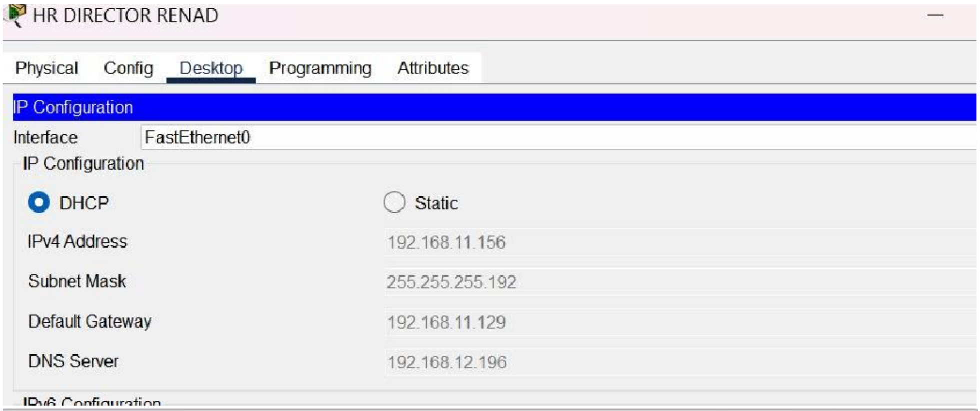
3. Step 3: Configured static routing between Router 1 and Router 2 to ensure inter-router communication.

```
Password:

B3R>ena
Password:
B3R#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
B3R(config)#
B3R(config)#int serial0/2/1
B3R(config-if)#ip add 10.10.10.37 255.255.252
                                        ^
% Invalid input detected at '^' marker.

B3R(config-if)#ip add 10.10.10.37 255.255.255.252
B3R(config-if)#no shutdown
B3R(config-if)#
```

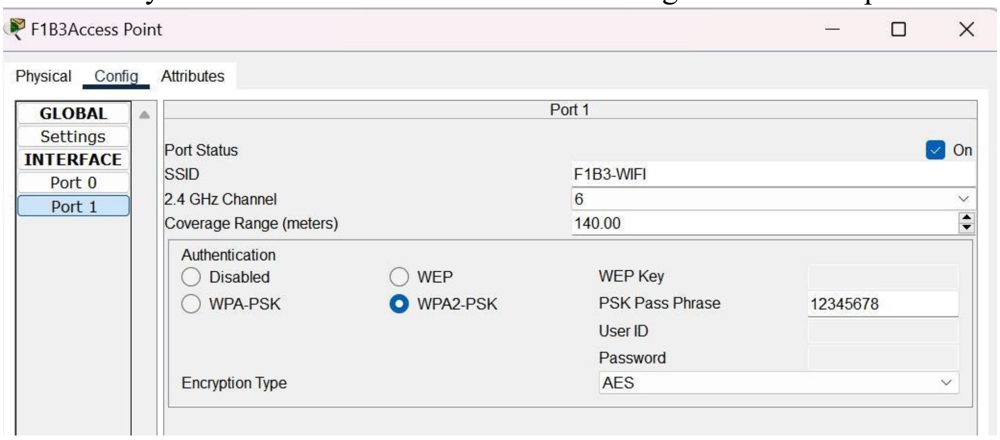
4. Step 4: Set up DHCP on Router 1 to provide dynamic IP addressing to connected devices. Configured IP pools and DNS settings.

```
B3R(config-if)#no shutdown
B3R(config-if)#ip dhcp pool LAN_POOL
B3R(dhcp-config)#network 10.10.10.0 255.255.255.0
B3R(dhcp-config)#default-roter 10.10.10.37
                            ^
% Invalid input detected at '^' marker.

B3R(dhcp-config)#default-router 10.10.10.37
B3R(dhcp-config)#dns-server 8.8.8.8
B3R(dhcp-config)#exit
B3R(config)#ip dhcp excluded-address 10.10.10.37
B3R(config)#
```

5. Step 5: Connected PCs and printers to the respective switches (Switch 1 for Router 1 and Switch 2 for Router 2). Assigned static IPs to devices where necessary.

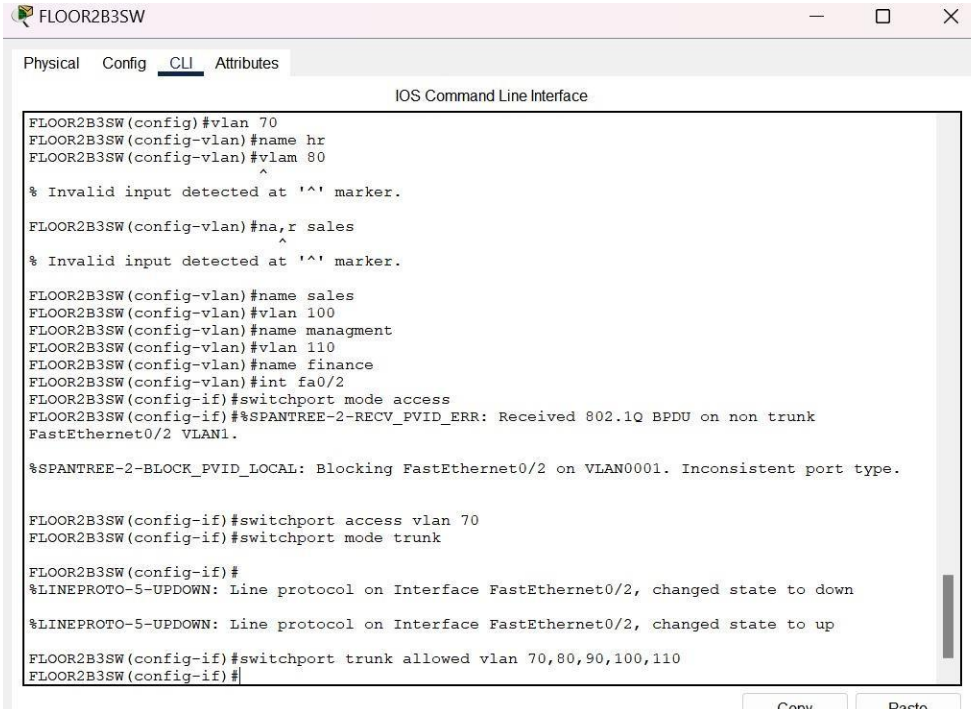


6. Step 6: Configured an Access Point connected to Switch 1, providing secure wireless connectivity. Wireless devices were connected using the SSID and password.

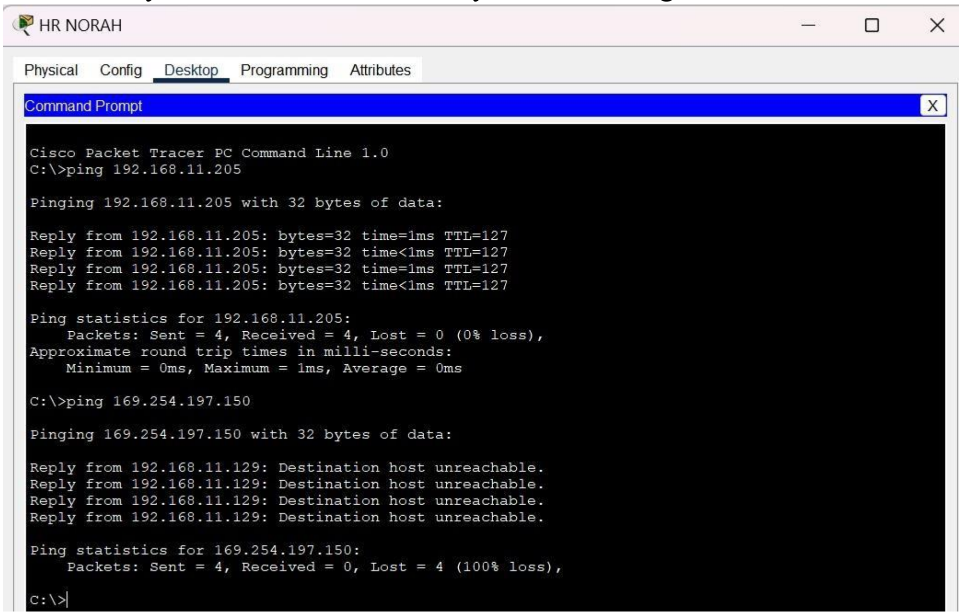


7. Step 7: Connected IP Phones to the switches and configured them for VoIP. Phones were assigned IPs and verified for connectivity.

8. Step 8: Implemented VLANs on the switches: VLAN 10 for Data, VLAN 20 for Voice. Assigned appropriate ports to each VLAN and set trunking for VLAN communication.



9. Step 9: Tested the network by pinging devices across subnets and verifying connectivity. DHCP was tested for dynamic IP assignment.



Conclusion

The project achieved complete network connectivity with effective IP addressing, routing, DHCP configuration, VLAN segmentation, and secure wireless access. All devices were successfully interconnected, and the network operated smoothly. This design demonstrates a comprehensive understanding of network design and configuration using Cisco Packet Tracer.